

Operating System Support

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- Operating System Objectives and Functions
- Types of Operating Systems

□ **Scheduling**

- Long-Term Scheduling
- Medium-Term Scheduling
- Short-Term Scheduling

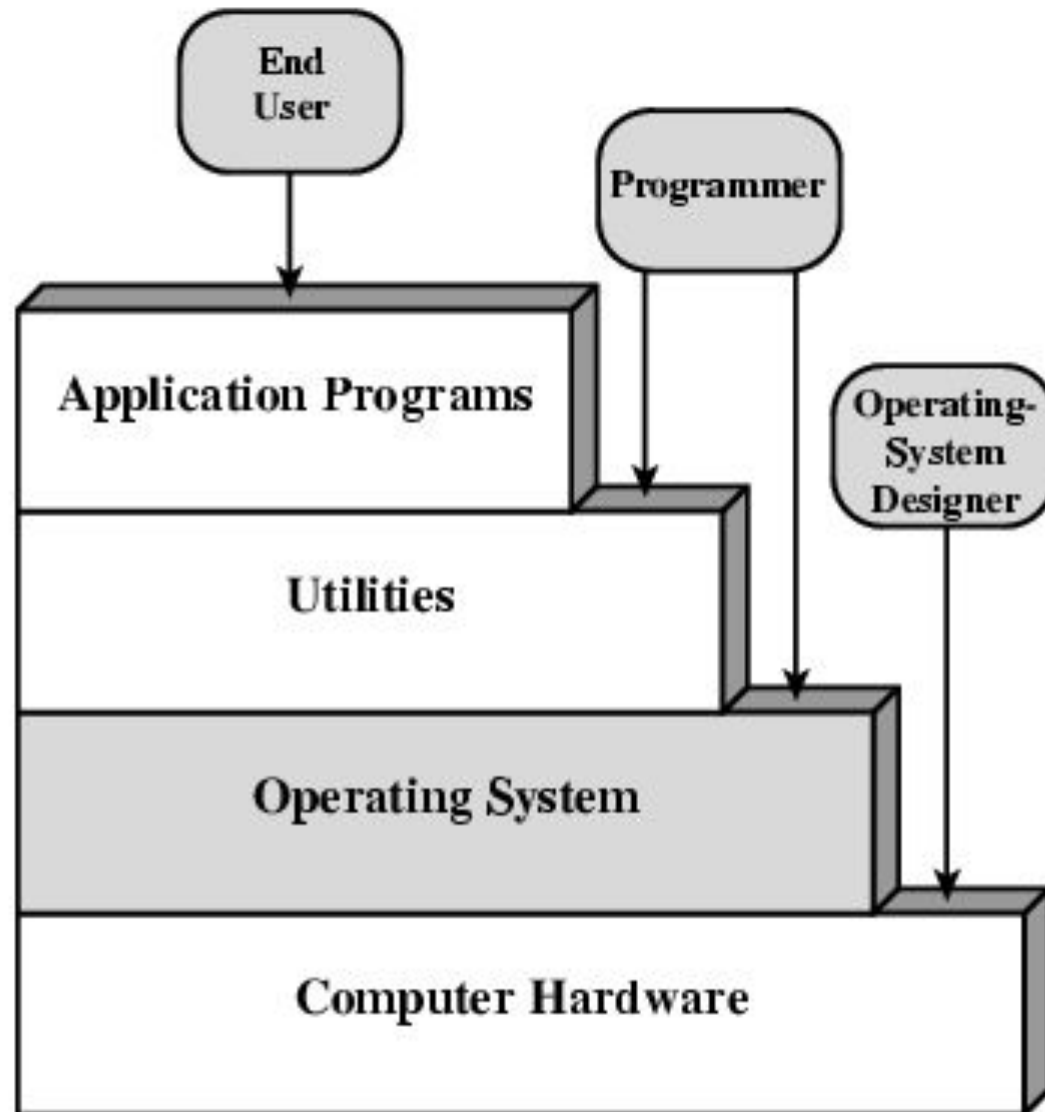
□ **Memory Management**

- Swapping
- Partitioning
- Paging
- Virtual Memory
- Segmentation

Objectives and Functions

- Convenience
 - Making the computer easier to use
- Efficiency
 - Allowing better use of computer resources

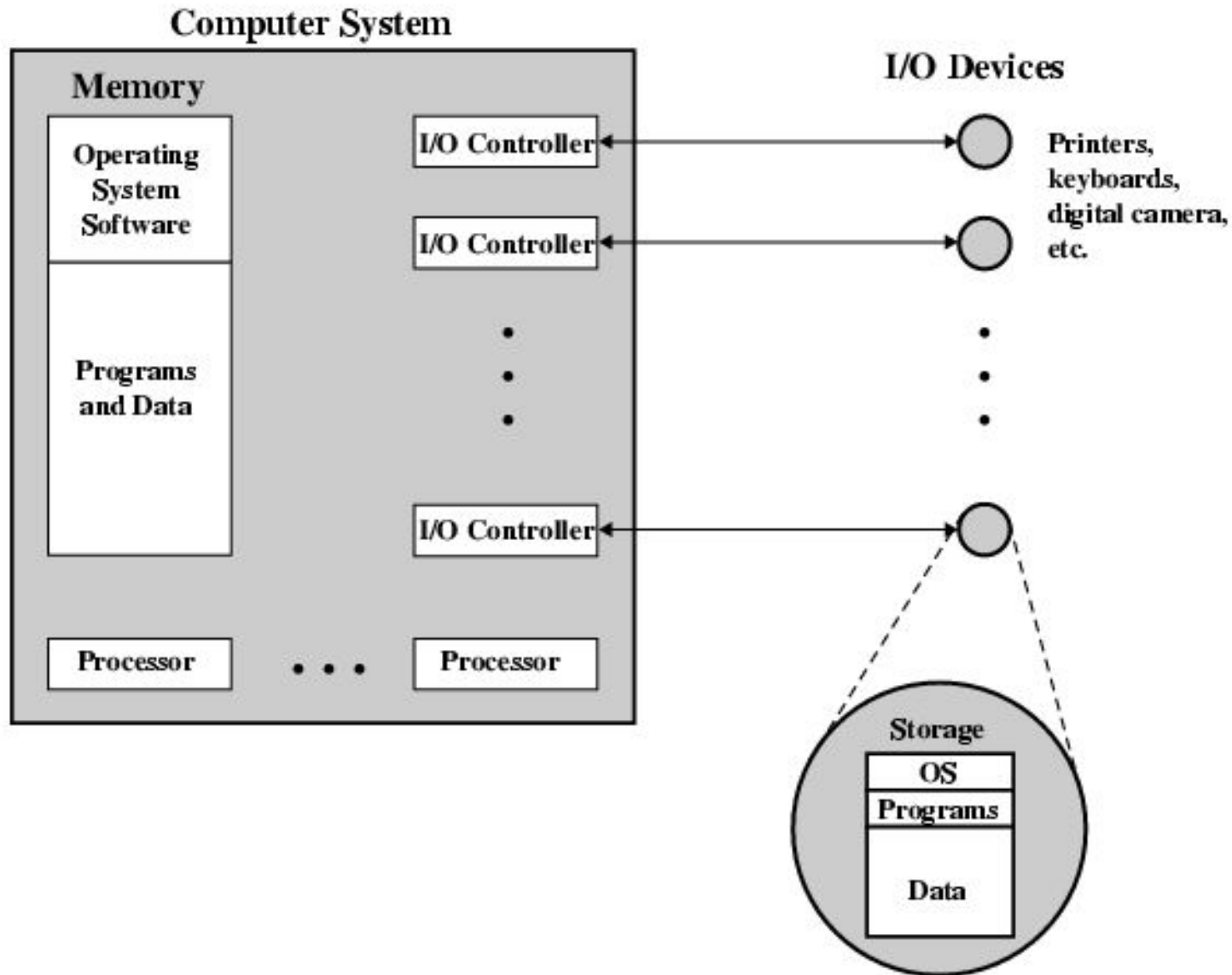
Layers and Views of a Computer System



Operating System Services

- Program creation
- Program execution
- Access to I/O devices
- Controlled access to files
- System access
- Error detection and response
- Accounting

O/S as a Resource Manager



Types of Operating System

- Interactive
- Batch
- Single program (Uni-programming)
- Multi-programming (Multi-tasking)

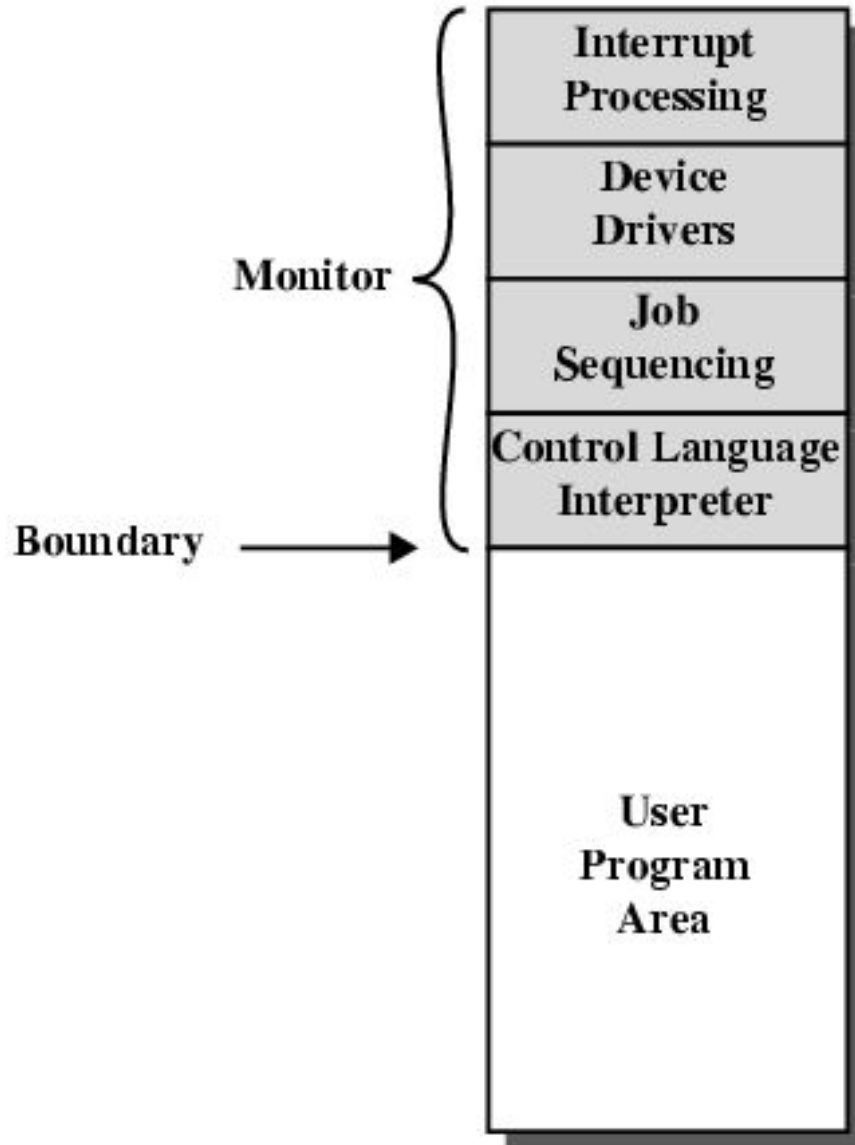
Early Systems

- Late 1940s to mid 1950s
- No Operating System
- Programs interact directly with hardware
- Two main problems:
 - Scheduling
 - Setup time

Simple Batch Systems

- Resident Monitor program
- Users submit jobs to operator
- Operator batches jobs
- Monitor controls sequence of events to process batch
- When one job is finished, control returns to Monitor which reads next job
- Monitor handles scheduling

Memory Layout for Resident Monitor



Job Control Language

- Instructions to Monitor
- Usually denoted by \$
- e.g.
 - \$JOB
 - \$FTN
 - ... Some Fortran instructions
 - \$LOAD
 - \$RUN
 - ... Some data
 - \$END

Desirable Hardware Features

- Memory protection
 - To protect the Monitor
- Timer
 - To prevent a job monopolizing the system
- Privileged instructions
 - Only executed by Monitor
 - e.g. I/O
- Interrupts
 - Allows for relinquishing and regaining control

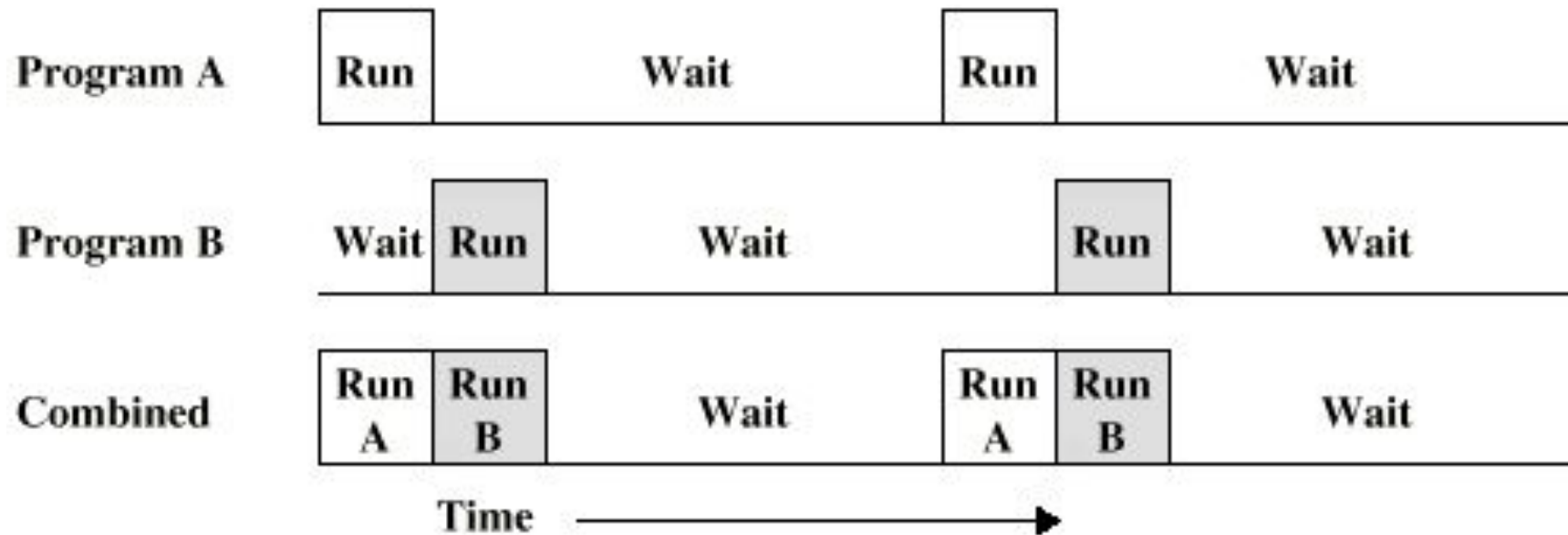
Multi-programmed Batch Systems

- I/O devices very slow
- When one program is waiting for I/O, another can use the CPU

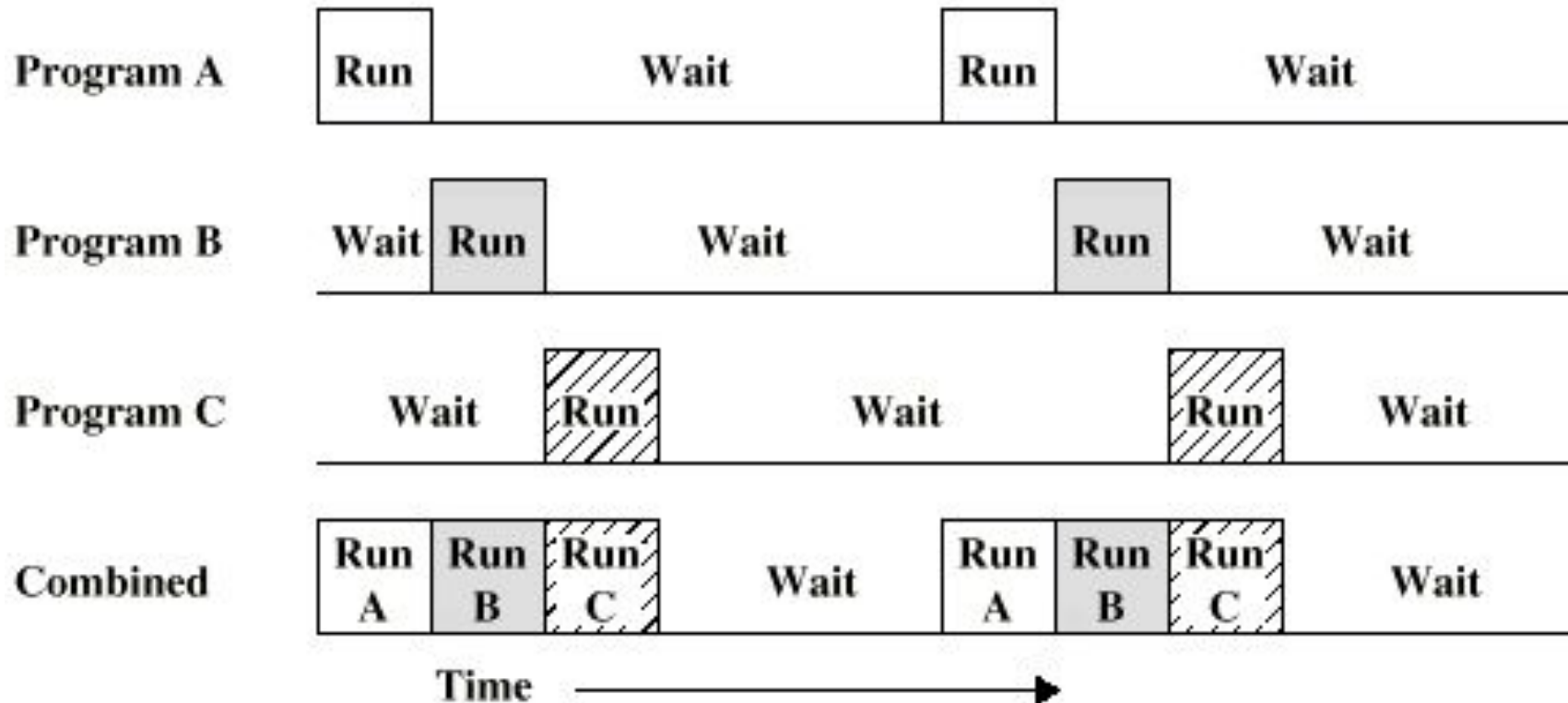
Single Program



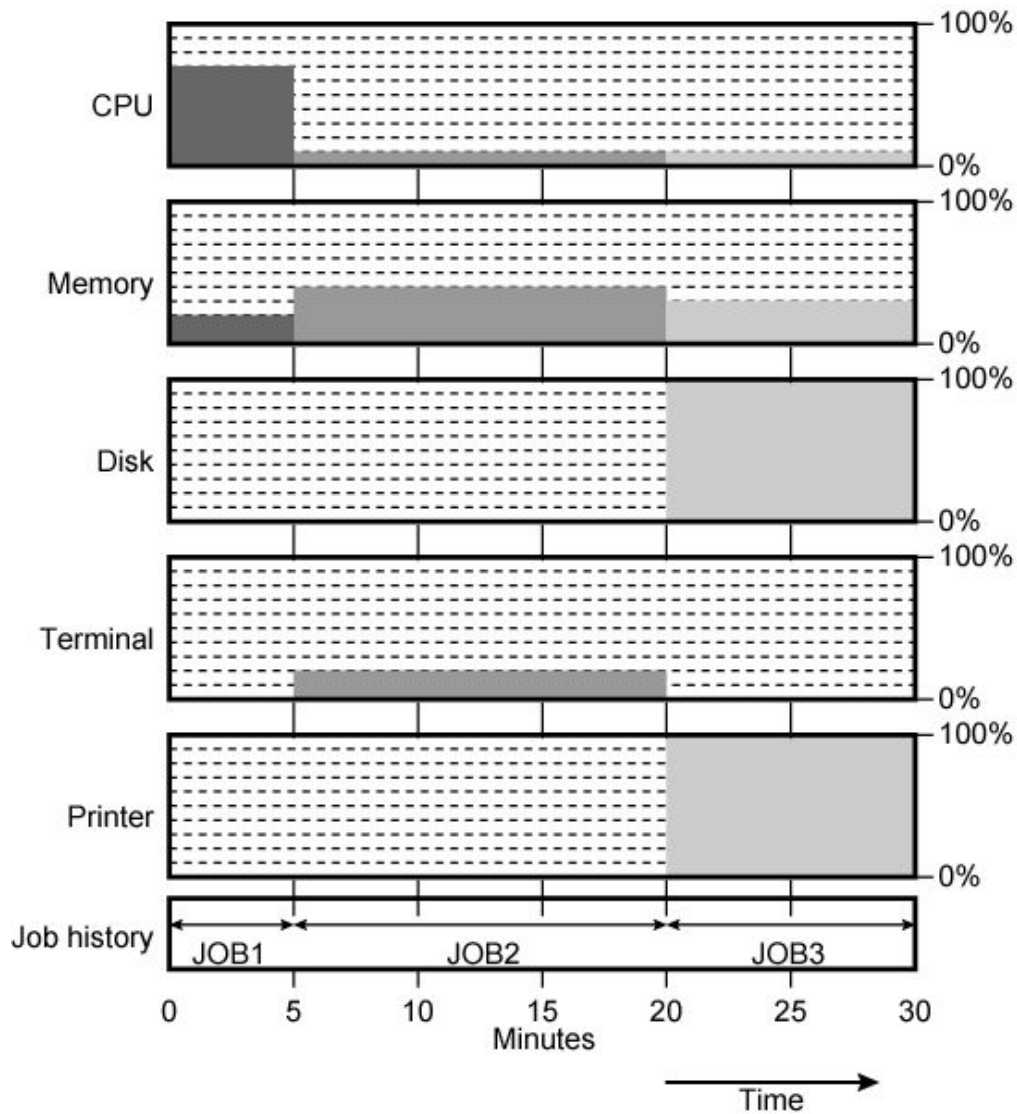
Multi-Programming with Two Programs



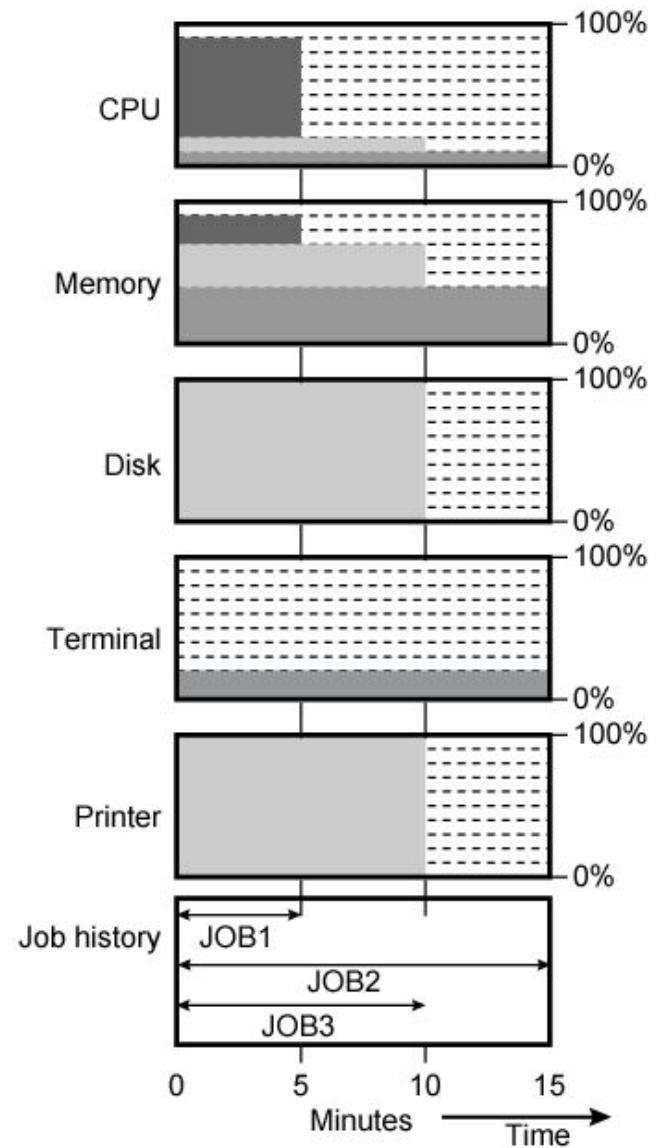
Multi-Programming with Three Programs



Utilization



(a) Uniprogramming



(b) Multiprogramming

Time Sharing Systems

- Allow users to interact directly with the computer
 - i.e. Interactive
- Multi-programming allows a number of users to interact with the computer

Scheduling

- Key to multi-programming
- Long term
- Medium term
- Short term
- I/O

Long Term Scheduling

- Determines which programs are submitted for processing
- i.e. controls the degree of multi-programming
- Once submitted, a job becomes a process for the short term scheduler
- (or it becomes a swapped out job for the medium term scheduler)

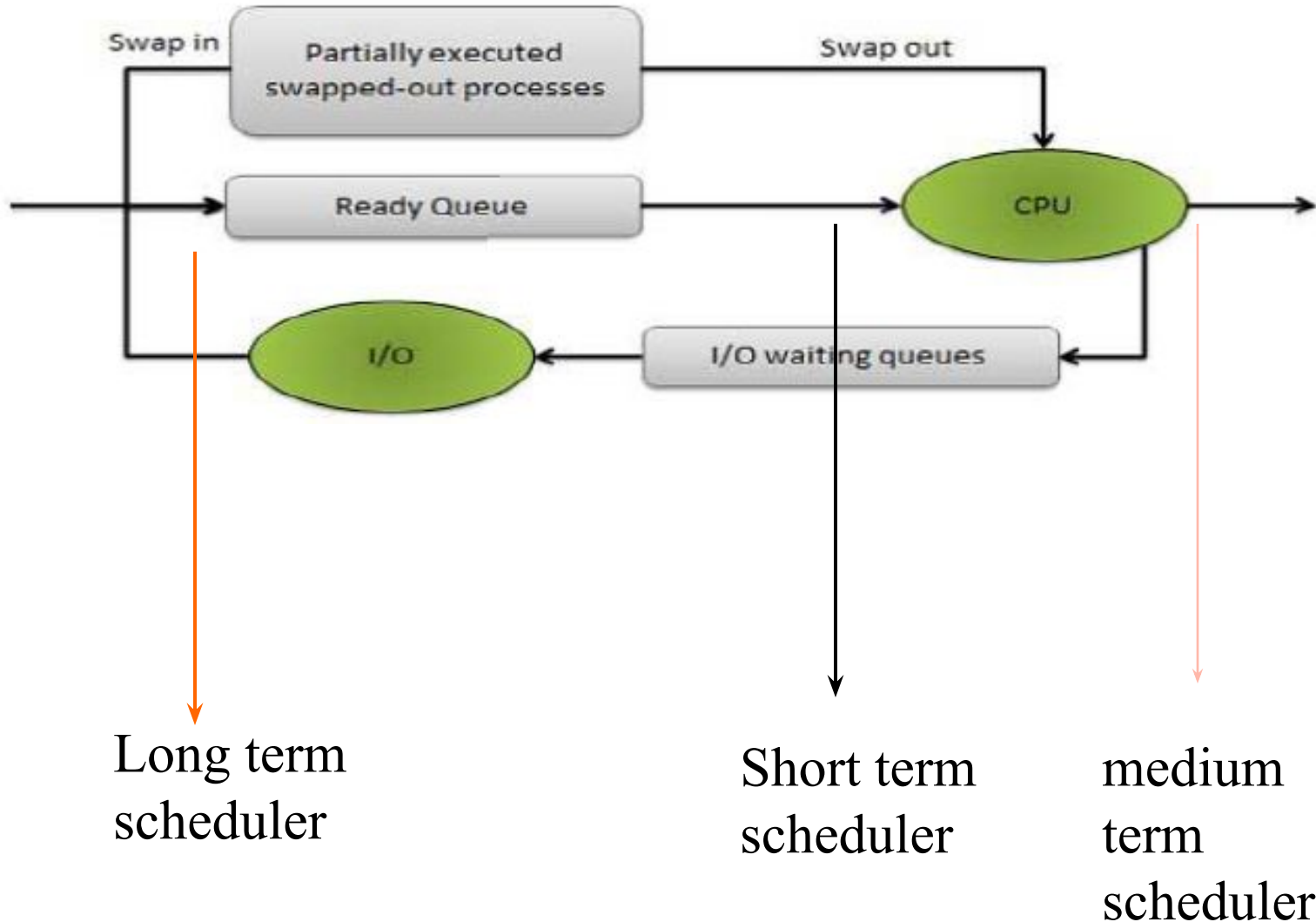
Medium Term Scheduling

- Part of the swapping function (later...)
- Usually based on the need to manage multi-programming
- If no virtual memory, memory management is also an issue

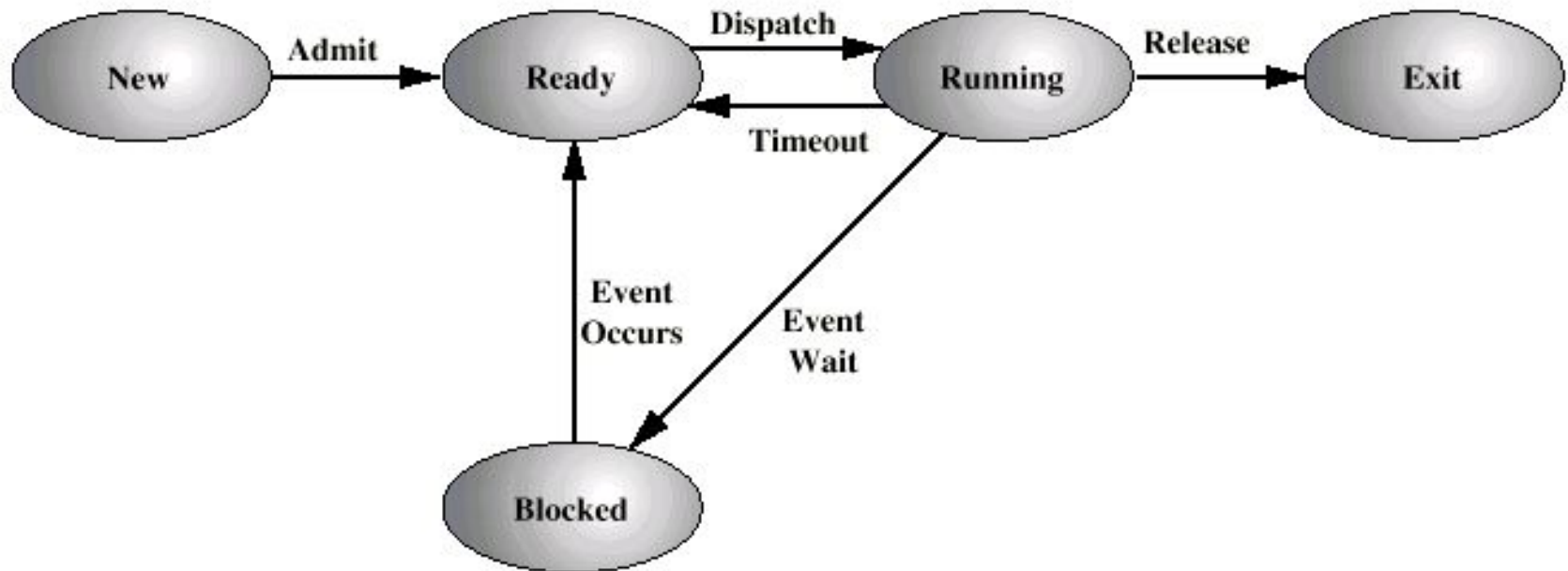
Short Term Scheduler

- Dispatcher
- Fine grained decisions of which job to execute next
- i.e. which job actually gets to use the processor in the next time slot

Cont.



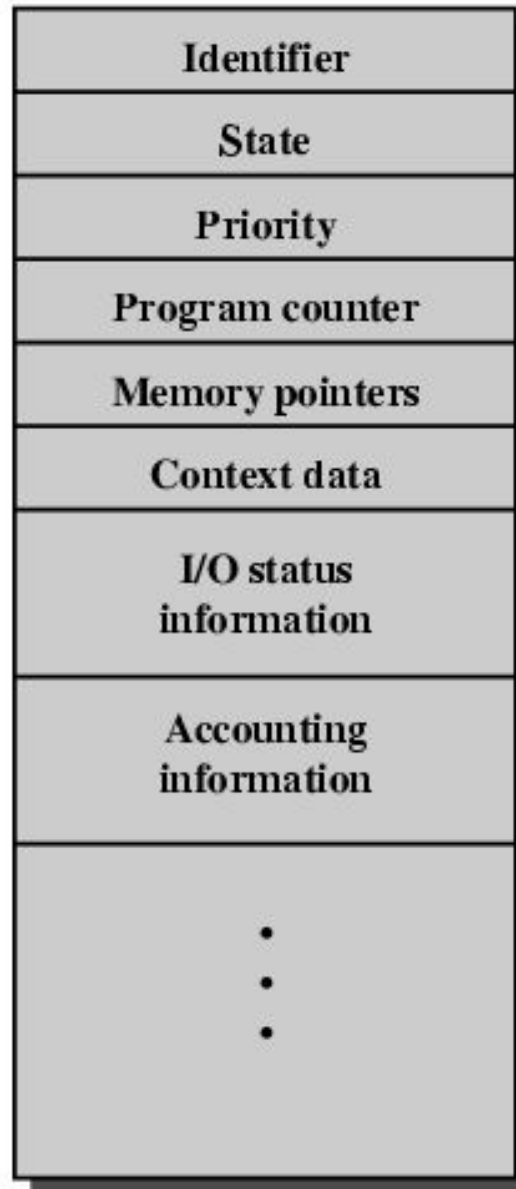
Five State Process Model



Process Control Block

- Identifier
- State
- Priority
- Program counter
- Memory pointers
- Context data
- I/O status
- Accounting information

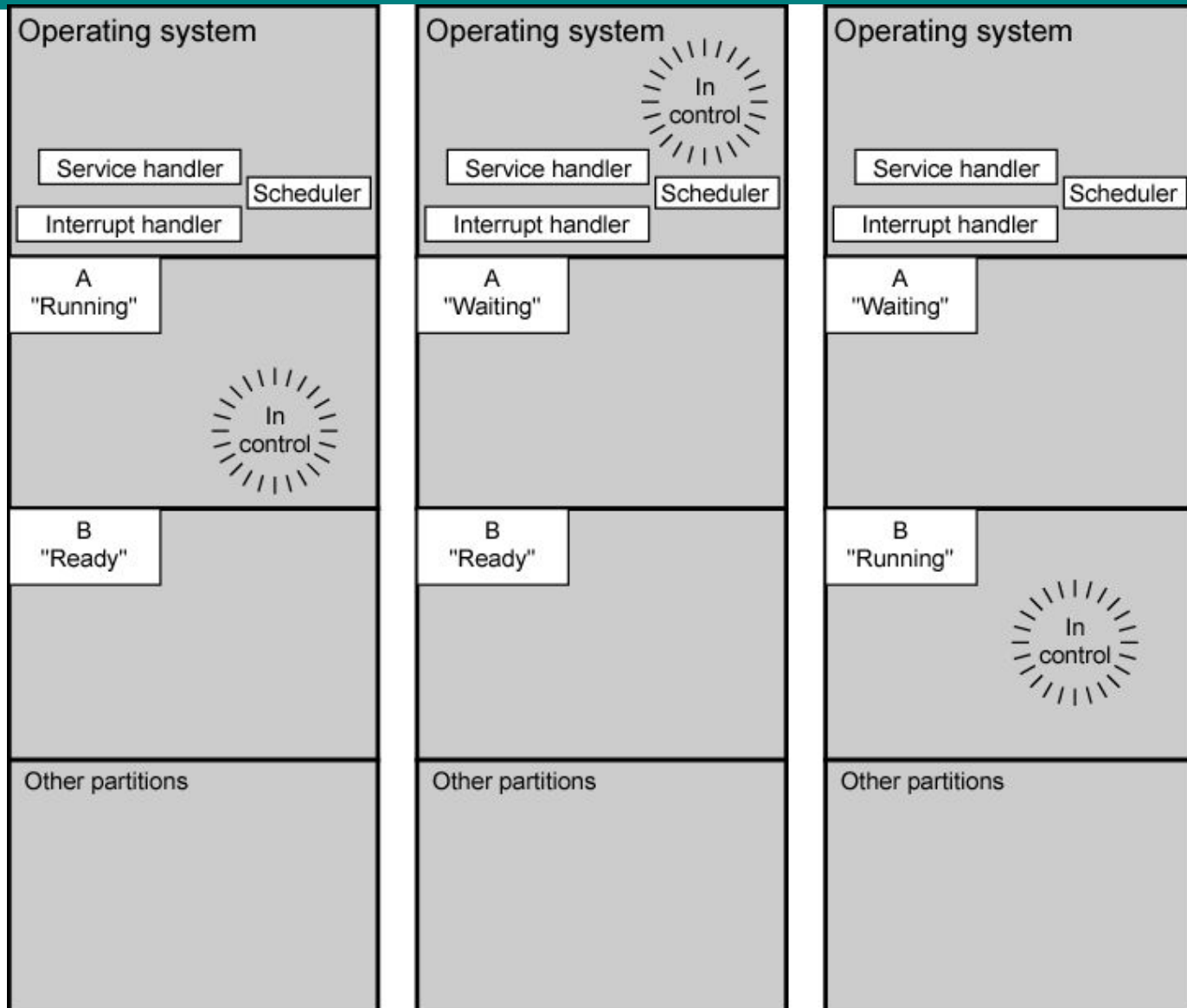
PCB Diagram



Cont.

S.N.	Information & Description
1	Pointer Pointer points to another process control block. Pointer is used for maintaining the scheduling list.
2	Process State Process state may be new, ready, running, waiting and so on.
3	Program Counter Program Counter indicates the address of the next instruction to be executed for this process.
4	CPU registers CPU registers include general purpose register, stack pointers, index registers and accumulators etc. number of register and type of register totally depends upon the computer architecture.
5	Memory management information This information may include the value of base and limit registers, the page tables, or the segment tables depending on the memory system used by the operating system. This information is useful for deallocating the memory when the process terminates.
6	Accounting information This information includes the amount of CPU and real time used, time limits, job or process numbers, account numbers etc.

Scheduling Example

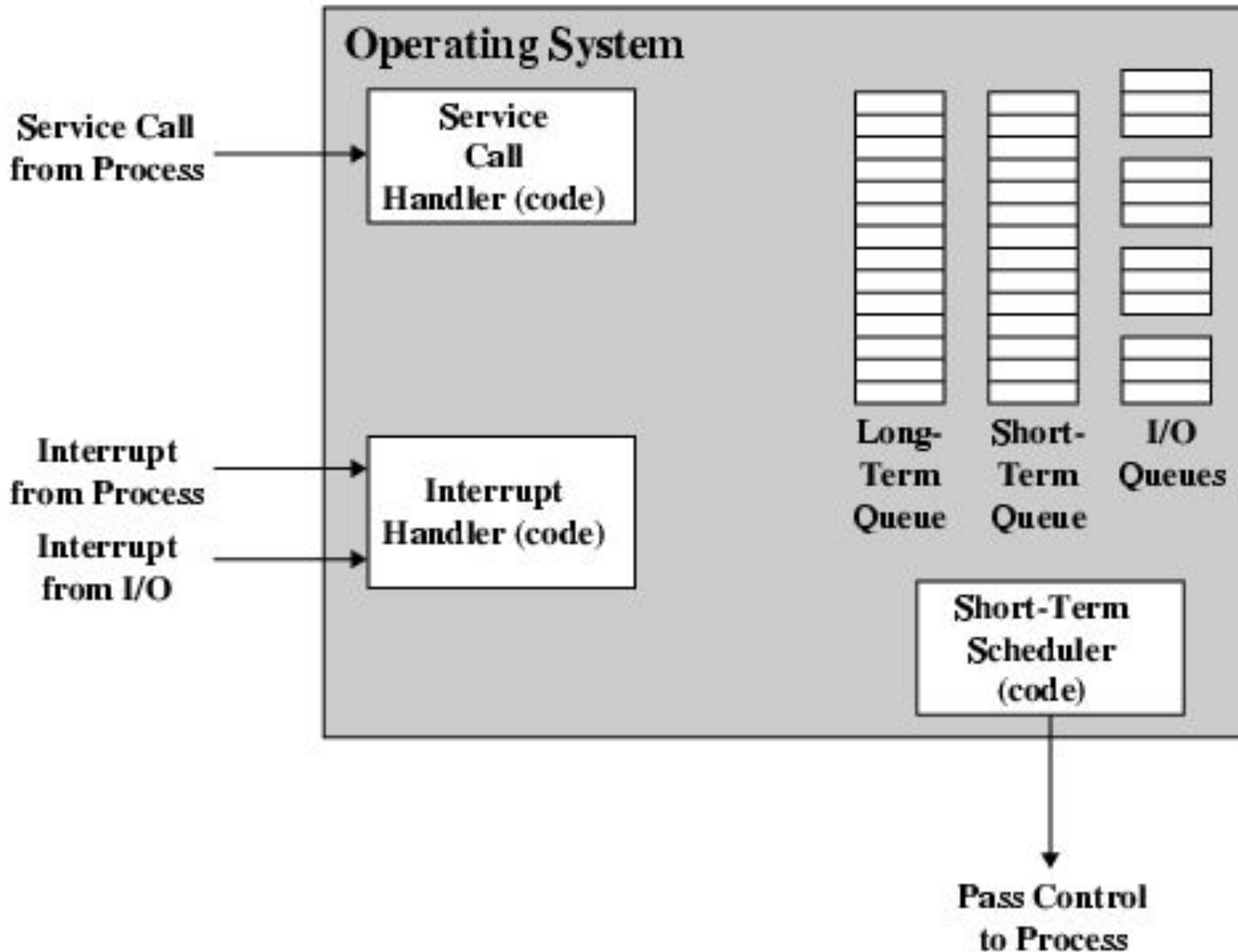


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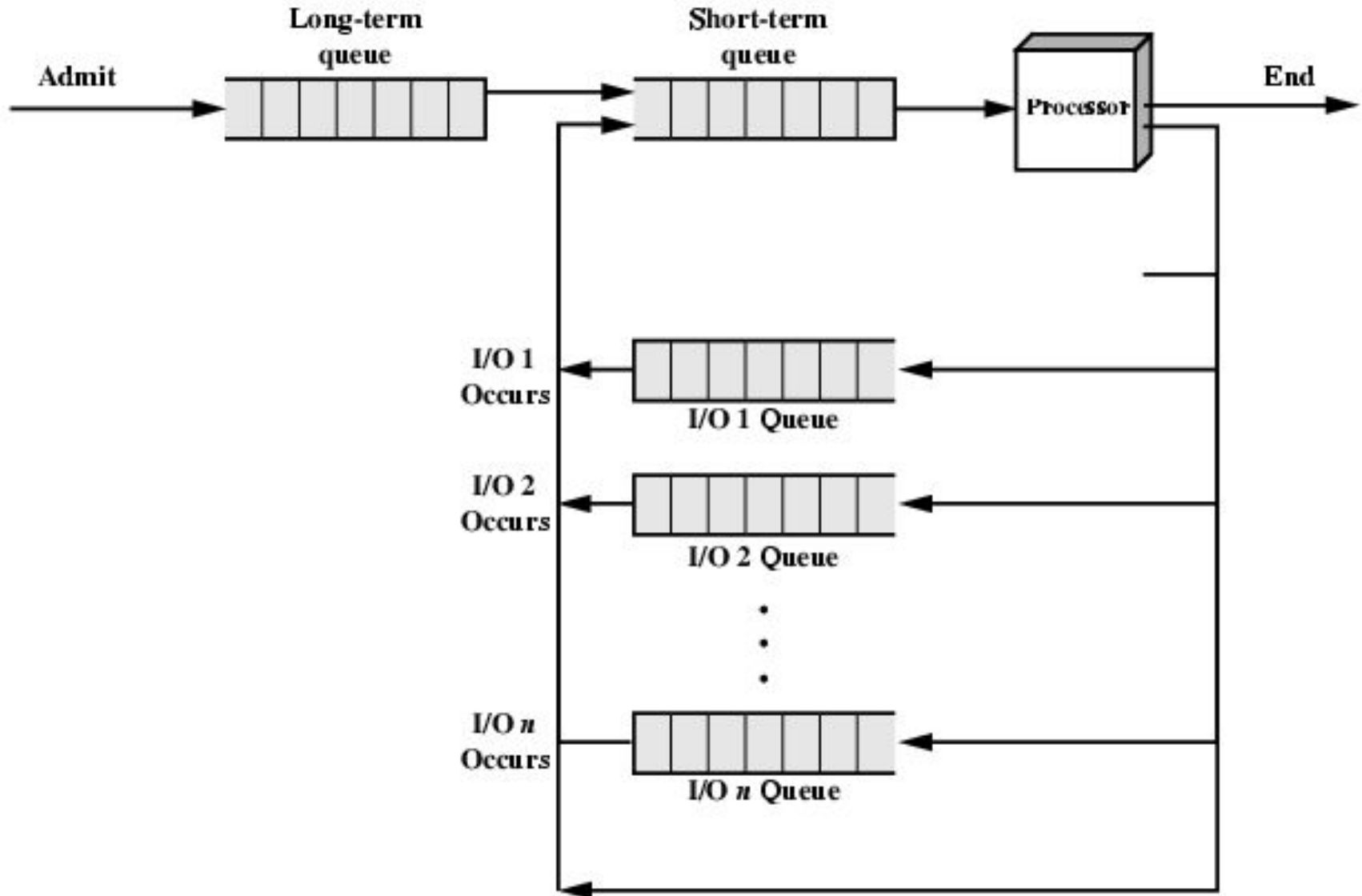
(b)

(c)

Key Elements of O/S



Process Scheduling



Memory Management

- Uni-program
 - Memory split into two
 - One for Operating System (monitor)
 - One for currently executing program
- Multi-program
 - “User” part is sub-divided and shared among active processes

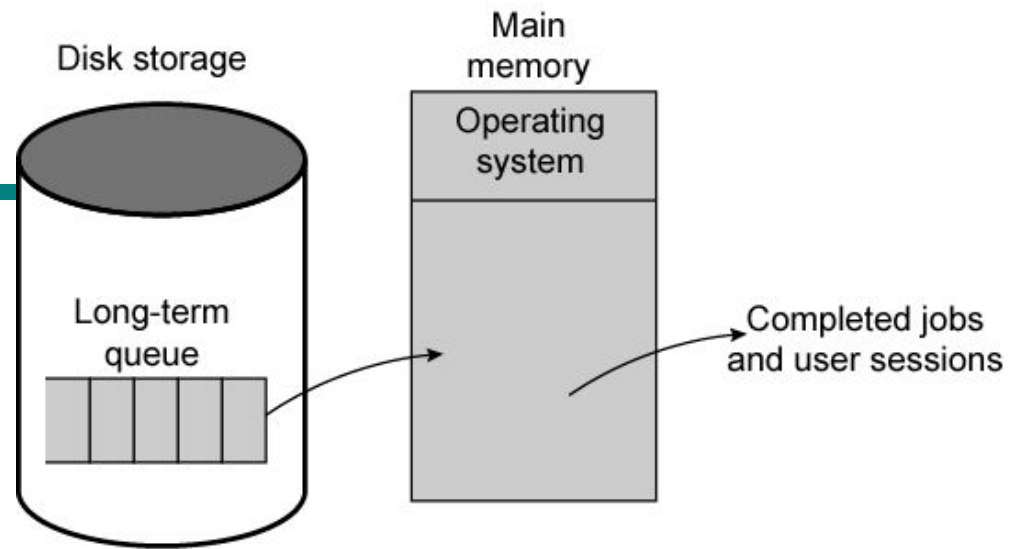
Swapping

- Problem: I/O is so slow compared with CPU that even in multi-programming system, CPU can be idle most of the time
- Solutions:
 - Increase main memory
 - Expensive
 - Leads to larger programs
 - Swapping

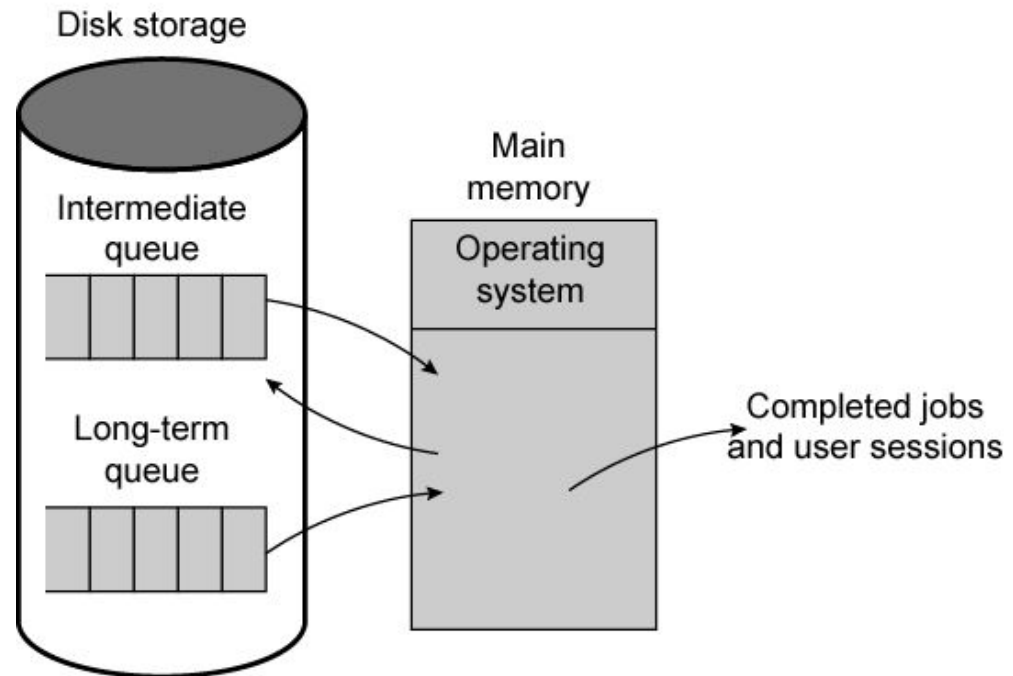
What is Swapping?

- Long term queue of processes stored on disk
- Processes “swapped” in as space becomes available
- As a process completes it is moved out of main memory
- If none of the processes in memory are ready (i.e. all I/O blocked)
 - Swap out a blocked process to intermediate queue
 - Swap in a ready process or a new process
 - But swapping is an I/O process...

Use of Swapping



(a) Simple job scheduling

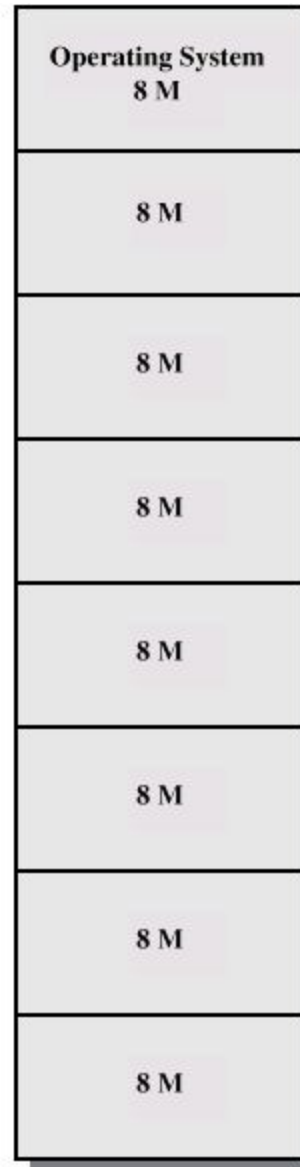


(b) Swapping

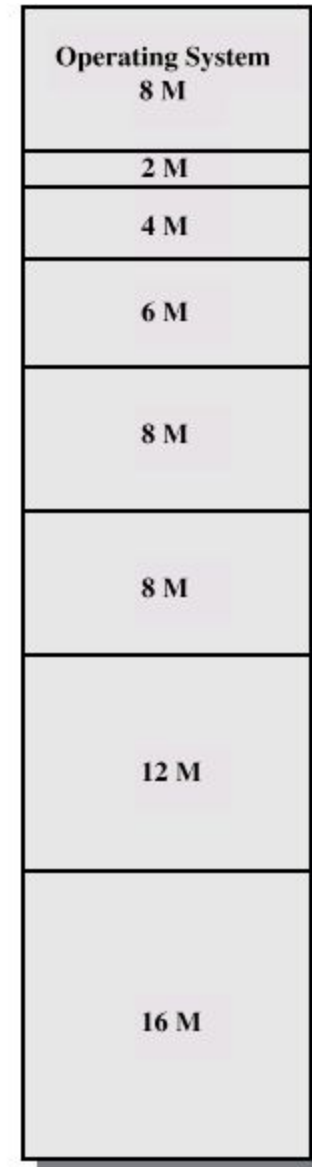
Partitioning

- Splitting memory into sections to allocate to processes (including Operating System)
- Fixed-sized partitions
 - May not be equal size
 - Process is fitted into smallest hole that will take it (best fit)
 - Some wasted memory
 - Leads to variable sized partitions

Fixed Partitioning



(a) Equal-size partitions



(b) Unequal-size partitions

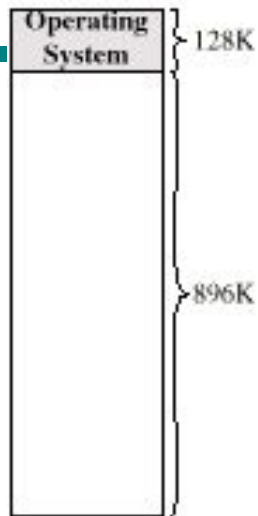
Variable Sized Partitions (1)

- Allocate exactly the required memory to a process
- This leads to a hole at the end of memory, too small to use
 - Only one small hole - less waste
- When all processes are blocked, swap out a process and bring in another
- New process may be smaller than swapped out process
- Another hole

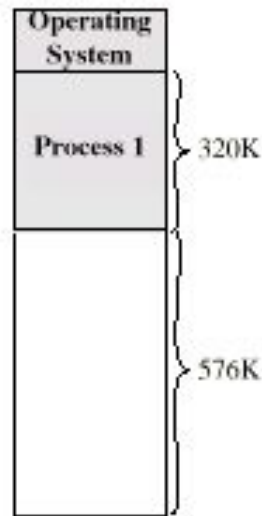
Variable Sized Partitions (2)

- Eventually have lots of holes (fragmentation)
- Solutions:
 - Coalesce - Join adjacent holes into one large hole
 - Compaction - From time to time go through memory and move all hole into one free block (c.f. disk de-fragmentation)

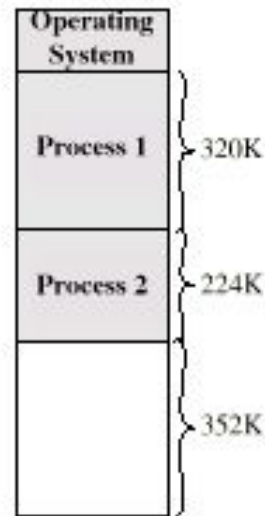
Effect of Dynamic Partitioning



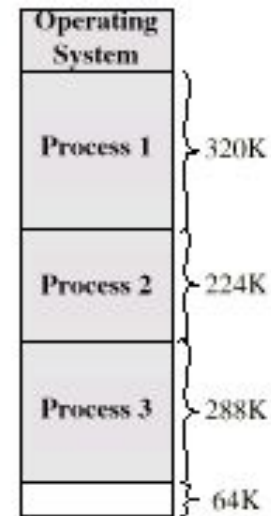
(a)



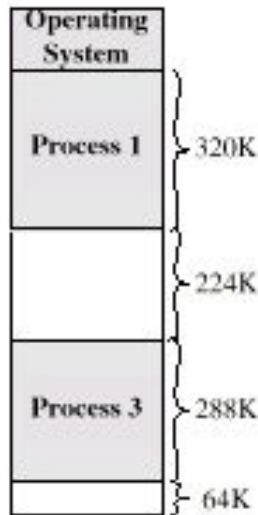
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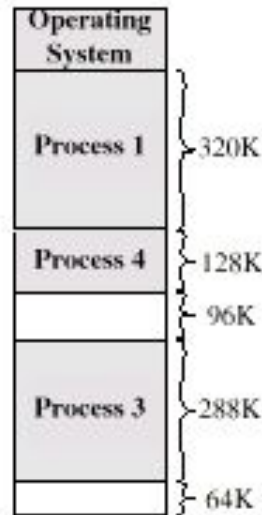
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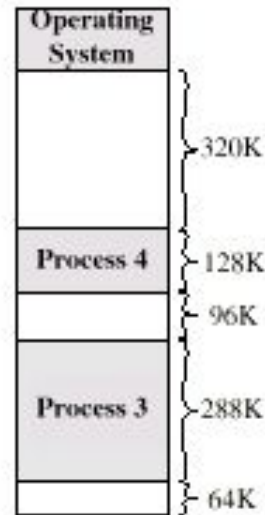
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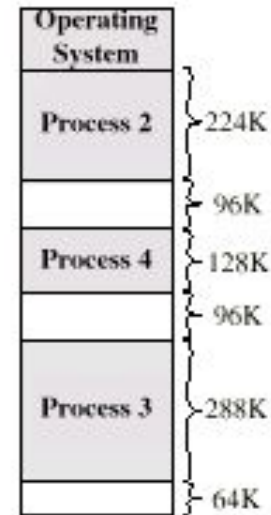
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(f)



(g)



(h)

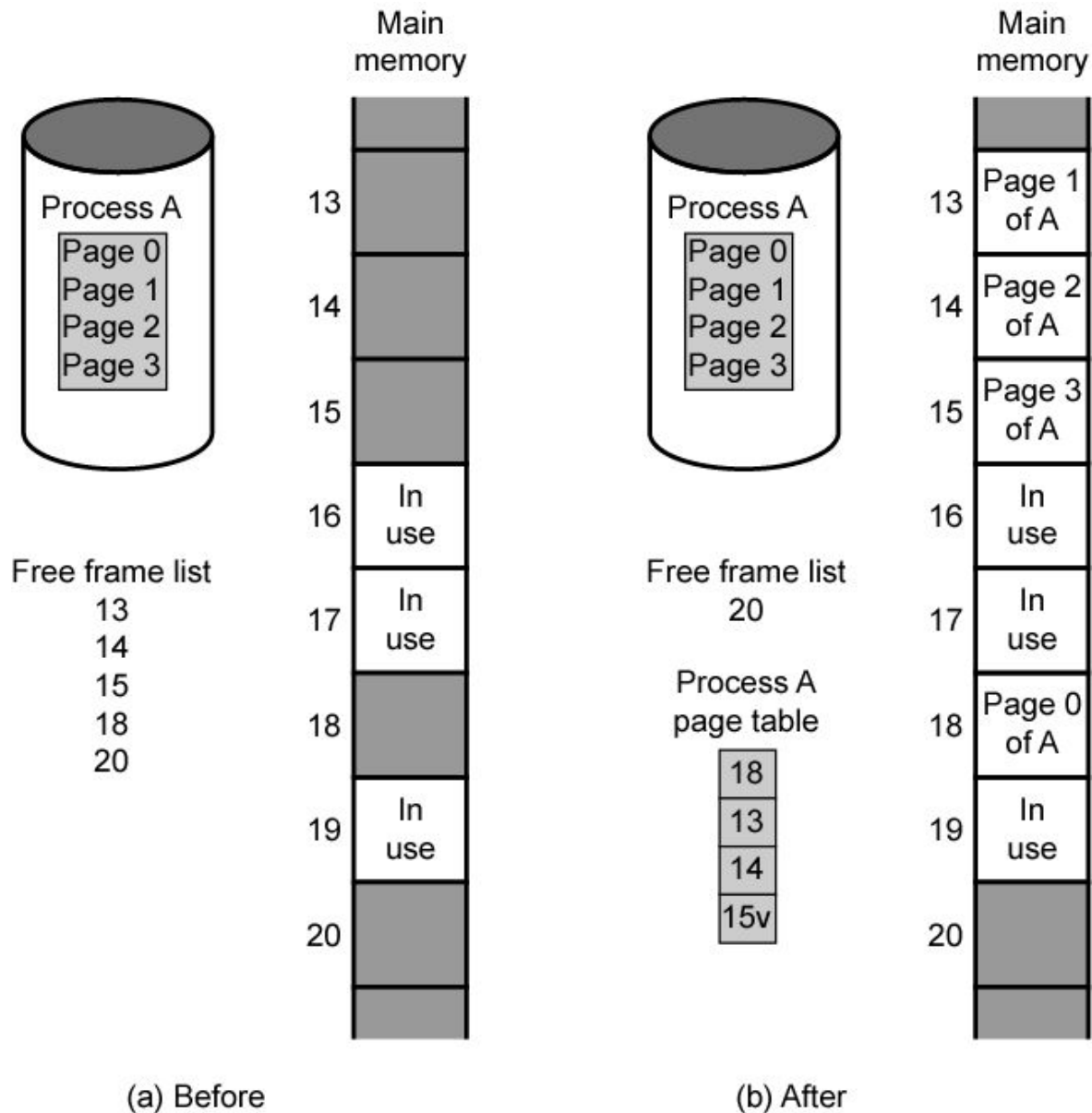
Relocation

- No guarantee that process will load into the same place in memory
- Instructions contain addresses
 - Locations of data
 - Addresses for instructions (branching)
- Logical address - relative to beginning of program
- Physical address - actual location in memory (this time)
- Automatic conversion using base address

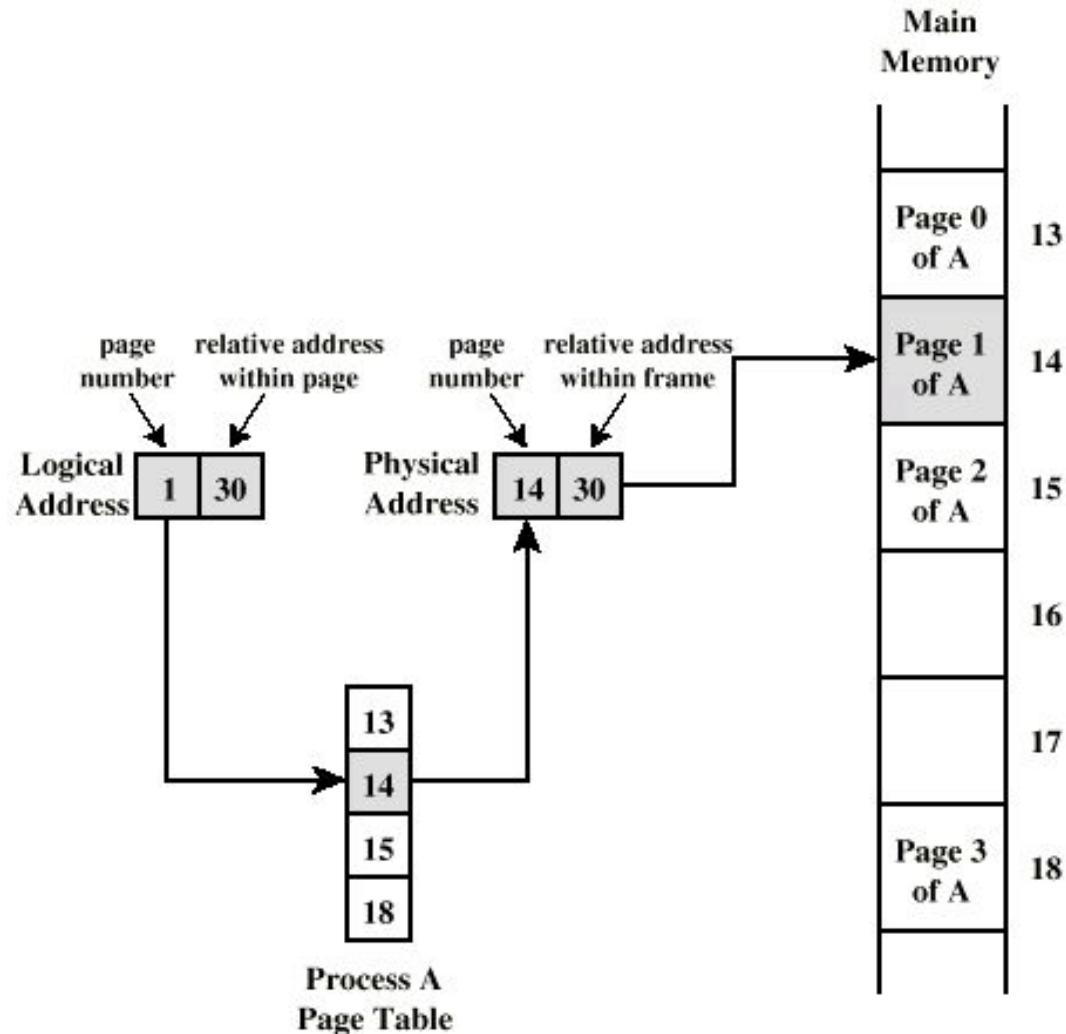
Paging

- Split memory into equal sized, small chunks -page frames
- Split programs (processes) into equal sized small chunks - pages
- Allocate the required number page frames to a process
- Operating System maintains list of free frames
- A process does not require contiguous page frames
- Use page table to keep track

Allocation of Free Frames



Logical and Physical Addresses - Paging



Virtual Memory

- Demand paging
 - Do not require all pages of a process in memory
 - Bring in pages as required
- Page fault
 - Required page is not in memory
 - Operating System must swap in required page
 - May need to swap out a page to make space
 - Select page to throw out based on recent history

Thrashing

- Too many processes in too little memory
- Operating System spends all its time swapping
- Little or no real work is done
- Disk light is on all the time

- Solutions
 - Good page replacement algorithms
 - Reduce number of processes running
 - Fit more memory

Bonus

- We do not need all of a process in memory for it to run
- We can swap in pages as required
- So - we can now run processes that are bigger than total memory available!
- Main memory is called real memory
- User/programmer sees much bigger memory - virtual memory

Required Reading

- Stallings chapter 8
- Stallings, W. [2004] *Operating Systems*, Pearson
- Loads of Web sites on Operating Systems

Thanks for attention